

Linear bubble plume model for hypolimnetic oxygenation: Full-scale validation and sensitivity analysis

V. L. Singleton,¹ P. Gantzer,¹ and J. C. Little¹

Received 23 December 2005; revised 12 June 2006; accepted 21 November 2006; published 8 February 2007.

[1] An existing linear bubble plume model was improved, and data collected from a full-scale diffuser installed in Spring Hollow Reservoir, Virginia, were used to validate the model. The depth of maximum plume rise was simulated well for two of the three diffuser tests. Temperature predictions deviated from measured profiles near the maximum plume rise height, but predicted dissolved oxygen profiles compared very well with observations. A sensitivity analysis was performed. The gas flow rate had the greatest effect on predicted plume rise height and induced water flow rate, both of which were directly proportional to gas flow rate. Oxygen transfer within the hypolimnion was independent of all parameters except initial bubble radius and was inversely proportional for radii greater than approximately 1 mm. The results of this work suggest that plume dynamics and oxygen transfer can successfully be predicted for linear bubble plumes using the discrete-bubble approach.

Citation: Singleton, V. L., P. Gantzer, and J. C. Little (2007), Linear bubble plume model for hypolimnetic oxygenation: Full-scale validation and sensitivity analysis, *Water Resour. Res.*, 43, W02405, doi:10.1029/2005WR004836.

1. Introduction

[2] Bubble plumes are used in a variety of industrial and environmental applications including mixing in chemical reactors, stripping of dissolved gases, containment of spills, prevention of ice formation, protection of harbors from damaging waves [Fanneløp *et al.*, 1991], and destratification of lakes and reservoirs [Schladow, 1992]. In addition to airlift aerators [Burris *et al.*, 2002] and Speece cones [McGinnis and Little, 1998], bubble plumes are commonly used for hypolimnetic aeration and oxygenation, which preserves stratification of water bodies while adding oxygen to the deepest layer. Hypolimnetic anoxia negatively affects the drinking water treatment process, cold-water fisheries, and water quality downstream of hydropower reservoirs. In the United States, releases from hydropower reservoirs typically must comply with state water quality criteria for minimum dissolved oxygen (DO) concentrations [Peterson *et al.*, 2003]. Oxygen depletion may lead to increases in hydrogen sulfide, ammonia, and phosphorus and the release of reduced iron and manganese from the sediments. Hydrogen sulfide, iron, and manganese in drinking water usually require additional treatment [Cooke and Carlson, 1989]. Finally, hypoxia can affect sex differentiation and development in fish, resulting in male-dominated populations with reduced reproductive success [Shang *et al.*, 2006].

[3] A bubble plume model to predict oxygen transfer from linear diffuser systems was presented by McGinnis *et al.* [2001], based on the model for a circular diffuser developed earlier by Wüest *et al.* [1992]. While several

models for point-source or circular bubble plumes have been proposed [Asaeda and Imberger, 1993; Brevik and Kluge, 1999; Ditmars and Cederwall, 1974; Fanneløp and Sjøen, 1980; Johansen, 2000; Kobus, 1968; McDougall, 1978; Milgram, 1983; Rayyan and Speece, 1977; Sahoo and Luketina, 2003; Schladow, 1992; Wüest *et al.*, 1992; Zheng *et al.*, 2002], less work has been conducted on linear (also referred to as line, two-dimensional, or planar) bubble plumes. Kobus [1968] developed one of the first analytical models for linear bubble plumes, which uses an empirical correlation to calculate buoyancy flux. Ditmars and Cederwall [1974] presented a model similar to that of Kobus [1968] but included bubble slip velocity. Brevik [1977] proposed a phenomenological theory for two-dimensional bubble plumes comparable with that of Ditmars and Cederwall [1974], except that kinetic energy was used to predict entrainment. Wilkinson [1979] proposed that full-scale linear plumes could be characterized by a Weber number. Lareshen and Rowe [1987] presented a model for two-dimensional bubble plumes in which plume spreading, entrainment, and momentum amplification were assumed to be functions of the plume Weber number and empirical constants. Fanneløp *et al.* [1991] developed a model for linear plumes in shallow water and studied the resulting surface currents and recirculation cells. Last, Brevik and Kluge [1999] expanded an existing model for linear bubble plumes to account for vertical turbulence. Although much insight into plume dynamics was gained, none of these models for linear or two-dimensional bubble plumes accounted for ambient stratification or gas transfer. The first linear bubble plume model to include gas transfer was presented by McGinnis *et al.* [2001], who converted the circular bubble plume model of Wüest *et al.* [1992] to linear geometry. The incorporation of gas transfer is critical because the rapid dissolution rate of oxygen, and nitrogen

¹Department of Civil and Environmental Engineering, Virginia Polytechnic Institute and State University, Blacksburg, Virginia, USA.

when compressed air is used, strongly influences the buoyancy of the plume [Wüest *et al.*, 1992]. Gas transfer is especially important in deep water bodies and for weak plumes because the increased contact time allows greater gas exchange. Last, the prediction of oxygen addition from hypolimnetic oxygenation systems is facilitated. Despite the usefulness of the linear bubble plume model, it has not yet been validated at full scale and over a range of operating conditions.

[4] Using extensive, high-spatial-resolution conductivity and temperature as a function of depth (CTD) transect data collected in Spring Hollow Reservoir (SHR), Virginia, during diffuser operation in 2003 and 2004, the performance of the linear bubble plume model is evaluated. The motivation for this work includes verification of model performance prior to use for design and investigation of critical model parameters through sensitivity analysis. Also, the accuracy of model predictions for depth of maximum plume rise (DMPR) and induced water flow rate should be assessed prior to coupling with lake/reservoir hydrodynamic and water quality models, such as CE-QUAL-W2 [McGinnis *et al.*, 2001]. In this paper an improved linear bubble plume model is presented, observations and model predictions are compared, and results of a sensitivity analysis are discussed.

2. Bubble Plumes in Stratified Water Bodies

[5] During hypolimnetic oxygenation with bubble plumes, compressed gas is continually supplied to diffusers, usually located immediately above the sediments, and is allowed to bubble freely. A gas-water plume mixture that is less dense than the ambient water is created, which causes the mixture to ascend through the hypolimnion. As the mixture rises, ambient water is entrained into the plume, and the plume width increases. The entrained fluid produces a double-plume structure, consisting of an inner core that contains the bubble-water mixture surrounded by an outer annulus that contains plume water relatively free of bubbles [McDougall, 1978]. As the outer annulus entrains stratified hypolimnetic water, the plume width increases and the density decreases. When the negative buoyancy of the entrained fluid exceeds the positive buoyancy imparted by the bubbles, the plume detains water at a rate nearly equal to that previously entrained [Lemckert and Imberger, 1993]. At this depth, the velocity of the relatively dense water within the plume decreases to zero, and the plume stops rising. The detaining plume water then forms an annular downward flow immediately outside the outer annulus of the upward flowing plume water. The detaining plume water entrains ambient water until a depth of neutral buoyancy is reached, where a horizontal intrusion is created into the hypolimnion [Asaeda and Imberger, 1993]. The undissolved bubbles remaining in the bubble-water mixture separate from the inner core flow and continue to rise to the surface, repeating the entire process.

3. Linear Bubble Plume Model

[6] The linear bubble plume model utilizes the discrete-bubble approach, which has also been applied to the airlift aerator and Speece cone and was recently reviewed in detail by Singleton and Little [2006]. The linear bubble plume

model is composed of horizontally integrated equations based on the conservation of mass, momentum, and heat. Eight flux equations are solved simultaneously to predict water flow rate, plume temperature, oxygen and nitrogen transfer and concentration, salinity, and plume rise height, given diffuser geometry and depth, applied gas flow rate, and initial bubble size (Tables 1 and 2). The model accounts for density stratification due to vertical temperature and salinity gradients. Entrainment is assumed to be proportional to the local (with respect to depth) plume water velocity and perimeter. Bubble size varies as the bubbles rise due to expansion and dissolution, and bubble slip velocity and gas transfer coefficients are functions of bubble radius [Wüest *et al.*, 1992]. Also, Henry's constants for oxygen and nitrogen are functions of temperature [Wüest *et al.*, 1992]. The bubble plume model equations were originally developed by Wüest *et al.* [1992] for circular geometry but were modified by McGinnis *et al.* [2001] for the linear geometry of the system installed in SHR. The equations that include the spreading coefficient (λ) were recently refined [Singleton and Little, 2005] to more accurately reflect the geometry of the plume at the ends of the linear diffuser (Tables 1 and 2), which is approximated in plan view as a long, thin rectangle. Additional refinements to the model of McGinnis *et al.* [2001], which are detailed in the following paragraphs, include use of a correlation to calculate initial bubble size [McGinnis and Little, 2002], correction of the entrainment coefficient (α) and λ for top-hat profiles, and use of a Froude number (Fr) to calculate initial water velocity [Fischer *et al.*, 1979; Wüest *et al.*, 1992]. Also, water quality profiles from the plume near-field, as opposed to the reservoir far-field, were used as boundary conditions [McGinnis *et al.*, 2004].

[7] The entrainment coefficient and λ were set at 0.11 and 0.93, respectively. These values were derived by Fanneløp *et al.* [1991] by fitting Gaussian profiles to laboratory data and were modified for the top-hat profile assumption of the model using [Fanneløp and Sjøen, 1980]

$$\alpha_T = \sqrt{2} \alpha_G \quad (1)$$

$$\lambda_T = \sqrt{\frac{\lambda_G^2 + 1}{2}}, \quad (2)$$

where the subscripts T and G refer to top-hat and Gaussian profiles, respectively. For simplicity, top-hat or uniform profiles are assumed for water velocity, temperature, salinity, dissolved and gaseous constituents, and bubble velocity [Wüest *et al.*, 1992]. Other model assumptions are as follows: (1) The linear plume width W for temperature and dissolved constituents is equal to the width of the plume velocity profile, whereas the bubbles are confined to an inner core of width λW ($\lambda < 1$); (2) ambient currents are negligible; (3) the diffuser produces bubbles at a constant rate and uniform size that are evenly distributed over the cross section of initial width λW_0 ; (4) bubble coalescence is neglected; (5) initial water properties of the plume are those of ambient water at the diffuser depth; and (6) exchange of gases other than oxygen and nitrogen is not considered.

[8] The model predictions are strongly dependent on the initial plume conditions and the plume boundary conditions

Table 1. Key Variables of the Linear Bubble Plume Model^a

Variable	Formula	Units
Entrainment factor	$E = 2(L + W)\alpha v$	m ² /s
Plume water volume flux	$Q = LWv$	m ³ /s
Momentum flux	$M = LWv^2$	m ⁴ /s ²
Temperature flux	$F_T = QT_p$	°C m ³ /s
Dissolved solids flux	$F_s = QS\rho_w$	kg/s
Dissolved O ₂ and N ₂ fluxes	$F_{D_i} = QC_i$	mol/s
Gaseous O ₂ and N ₂ fluxes	$F_{G_i} = \lambda W[L - W(1 - \lambda)](v + v_b)v_i$	mol/s

^aRevised from McGinnis *et al.* [2001].

of temperature, dissolved oxygen, and salinity. Initial conditions for the bubble plume model were determined as detailed by Wüest *et al.* [1992], except for the following deviations. Initial bubble size is calculated using the correlation developed by McGinnis and Little [2002] for the type of linear diffuser installed in SHR:

$$d_{3,2} = 1.12 + 0.938q. \quad (3)$$

The correlation was determined using measured Sauter-mean bubble diameter ($d_{3,2}$) values of 1.1–2.2 mm collected over actual unit gas flow rates q of 0.08–0.88 m²/h at the diffuser.

[9] For the circular bubble plume model, Wüest *et al.* [1992] proposed that the induced vertical water velocity v at the diffuser depth is equivalent to the initial plume water velocity. To estimate the initial velocity, Wüest *et al.* [1992] defined a densimetric Fr and utilized a relationship between the local Richardson number (Ri) and Fr derived by Fischer *et al.* [1979] for single-phase, round buoyant jets discharging vertically. The circular bubble plume model was recently validated by McGinnis *et al.* [2004], so a similar procedure was used to determine the initial plume velocity for the linear bubble plume. Corresponding equations for Fr were derived for planar or linear plumes using relationships presented by Fischer *et al.* [1979] to obtain

$$Fr = Ri^{-\frac{3}{4}} \quad (4)$$

$$Fr = \frac{v}{\left[\lambda Wg(\rho_a - \rho_p)/\rho_p\right]^{\frac{1}{2}}}, \quad (5)$$

where g is gravitational acceleration, ρ_a is the ambient water density, and ρ_p is the bubble plume density. The local Ri for planar jets has a constant value of 0.735 at distances from

the source where the flow is more like a plume [Fischer *et al.*, 1979]. Consequently, Fr for a planar or linear plume is equal to 1.26, except close to the source. Wüest *et al.* [1992] assumed that the bubble slip velocity near the source was relatively low, so the initial Froude number (Fr_o) should be equal to the value for a single-phase plume. Unlike the circular plume model, the Fr profiles predicted by the linear bubble plume model continually increase with depth for the diffuser installed in SHR (not shown). For $Fr_o \leq 2.0$, the plume velocity initially increases with decreasing depth immediately above the linear diffuser (Figure 1a). This effect was also predicted by Fanneløp and Webber [2003] for buoyant plumes rising from areal sources, where a point of maximum velocity occurred above the source. Additionally, the plume neck (point of minimum radius or width) is always below the point of maximum velocity in the plume [Fanneløp and Webber, 2003]. A neck is not predicted for the linear bubble plume when $Fr_o = 2.0$ (Figure 1b). However, necking or contraction will only occur when the momentum immediately above the source is relatively low and/or the entrainment coefficient is relatively low [Fanneløp and Webber, 2003]. Also, Wüest *et al.* [1992] reasoned that a plume from an open source (diffuser above sediments) may not contract because initially entrained water is not obstructed. Therefore Fr_o for an open source will likely be higher than that for a closed source (diffuser resting on sediments). Because the linear diffuser in SHR is an open source (Figure 2), Fr_o for the linear bubble plume was assumed to be 2.0, and a sensitivity analysis was conducted to determine the effect of varying Fr_o .

[10] The differential flux equations of the linear bubble plume model (Table 2) were solved numerically using the fourth-order Runge-Kutta method. Further information on the general solution procedure, equations of state, and model assumptions is provided by Wüest *et al.* [1992] and McGinnis *et al.* [2004]. The model calculations are only valid over the plume rise height, up to the DMPR. When the plume stops rising, a secondary plume may form above as bubbles that are not completely dissolved continue to rise [Asaeda and Imberger, 1993; McDougall, 1978; Schladow, 1992]. This phenomenon can occur when a bubble plume is released into strong density stratification.

4. Application to Spring Hollow Reservoir, Virginia

4.1. Field Data Collection

[11] To fully evaluate the linear bubble plume model, experimental data for boundary conditions, rise height, and in-plume constituent profiles are required. Testing was

Table 2. Nonlinear Differential Flux Equations of the Linear Bubble Plume Model^a

Flux	Equation
Water volume	$dQ/dz = E$
Momentum	$dM/dz = [(\rho_a - \rho_w)/\rho_p]gLW + [(\rho_w - \rho_p)/\rho_p]g\lambda W[L - W(1 - \lambda)]$
Temperature	$dF_T/dz = ET_a$
Salinity	$dF_s/dz = E\rho_a S_a$
Dissolved gas	$dF_{D_i}/dz = EC_a + [4\pi r^2 N/(v + v_b)]K_L(H_i P_i - C_i)$
Gas	$dF_{G_i}/dz = -[4\pi r^2 N/(v + v_b)]K_L(H_i P_i - C_i)$

^aRevised from McGinnis *et al.* [2001].

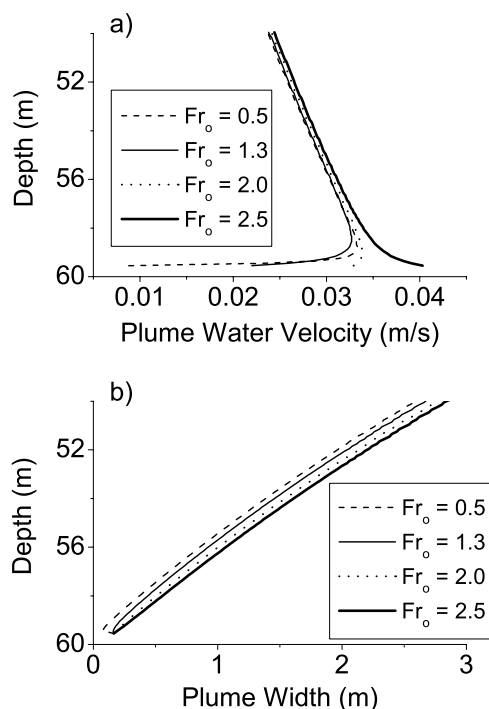


Figure 1. Effect of initial Froude number (Fr_o) on predictions of plume water velocity and plume width using linear bubble plume model.

conducted using a full-scale linear diffuser (Figure 2) installed in Spring Hollow Reservoir, Virginia (Figure 3). Constructed in 1995, SHR is a small monomictic, mesotrophic side-stream reservoir that is generally stratified from May to December. The reservoir is managed by the Western Virginia Water Authority and serves as one of the principle drinking water sources for Roanoke County. The water body has a maximum depth of 65 m and a maximum surface elevation of 431 m. The approximate surface area and volume are 0.54 km² and 12.4×10^6 m³, respectively. To prevent anoxia in the hypolimnion and the associated deterioration of raw water quality, a linear diffuser equipped with fine-bubble porous hoses was installed in 1997 (Figure 3). The 305-m-long diffuser can be supplied with compressed air or pure oxygen at various gas flow rates and is located in the deepest portion of the reservoir (368–372 m elevation). On the basis of an average surface elevation of 430 m, the depth of the diffuser during testing ranged from 58 to 62 m along its length. Table 3 provides the diffuser operating parameters in 2003 and 2004.

[12] Diffuser tests were performed in 2003 using compressed air (21% O₂) supplied at a high gas flow rate (45 Nm³/h average) during 29 June to 14 July and pure oxygen (97% O₂) supplied at a low flow rate (11 Nm³/h average) during 14–26 August. A third test was conducted in 2004 using pure oxygen, but at a higher gas flow rate (40 Nm³/h maximum) during 22 October to 5 November. In 2003, the diffuser quickly mixed the rather small hypolimnetic volume during both tests, and the water quality conditions on the dates of data collection (2 July and 17 August for compressed air and pure oxygen, respectively) were by that time relatively homogeneous as a result of plume-induced mixing. One of the primary objectives of the 2004

experiments was to maximize the plume signature in the hypolimnion and to increase confidence in the linear plume model validation under a different set of boundary conditions. In 2004, the data were therefore collected on 24 October, soon after start of diffuser operation. Additionally, the 2004 test was performed later in the stratified season to maximize the ambient DO and temperature gradients in the hypolimnion.

[13] To establish appropriate boundary conditions for the plume model, characterization of the plume near-field environment is necessary [McGinnis *et al.*, 2004]. Therefore the data collected included numerous high-spatial-resolution CTD (Sea-Bird model SBE 19plus; 4-Hz sampling rate) transects measured almost daily before, during, and after diffuser operation. The CTD profiler was also equipped with a DO probe (1.4-s response time measured at 20°C). Profiles were obtained laterally across the diffuser at 0.5-m increments for 0–10 m, 2-m increments for 10–20 m, and 5-m increments for 20–40 m from the centerline of the diffuser in both directions (Figures 3 and 4). (The diffuser centerline location is shifted to the left for 2004 (Figures 4e and 4f) because the diffuser was repositioned earlier in the year. Also, the operational length of the diffuser was decreased for that year (Table 3).)

[14] The geometry of SHR affects the extent and rate of circulation within the hypolimnion induced by the bubble plume. Because of the relatively small size of SHR, operation of the linear diffuser created uniform conditions below the thermocline within days after startup. While SHR bathymetry influences plume-induced mixing in the hypolimnion, the effect on short-term plume operation is negligible because the time that individual bubbles spend in the hypolimnion is of the order of minutes. The bubbles and resulting plume experience a pseudo steady state with respect to ambient conditions. Effects on data collection due to the ends of the linear diffuser were assumed to be negligible because the lateral profile location was over 150 m from a diffuser end (Figure 3), and the diffuser is designed to release a uniform gas flow along its length.

4.2. Observations and Model Validation

[15] Plume rise height, spreading, and constituent profiles predicted by the linear bubble plume model were compared with experimental observations. Critical model input parameters for the three test conditions (2 July 2003, 17 August 2003, and 23 October 2004) are shown in Table 3. The boundary conditions were obtained from averaged near-field lateral profiles [McGinnis *et al.*, 2004] (± 2 m and ± 1 m from plume centerline for 2003 and 2004, respectively) and differed significantly between 2003 and 2004 (Figure 5). (The presence of two thermoclines in 2003 is due to the pumped reservoir inflow that discharges at 396-m elevation, which corresponded to approximately 34-m depth during diffuser testing. The lower thermocline delineates the effective hypolimnion for the oxygenation system.)

[16] Measured contours of temperature and DO are shown in Figure 4, along with corresponding model predictions for plume width and the DMPR. The actual plume boundaries are not well defined in the contour plots, so comparison with predicted plume widths is difficult. The lack of distinct plume boundaries was due to the almost well mixed conditions in the hypolimnion as a result of diffuser

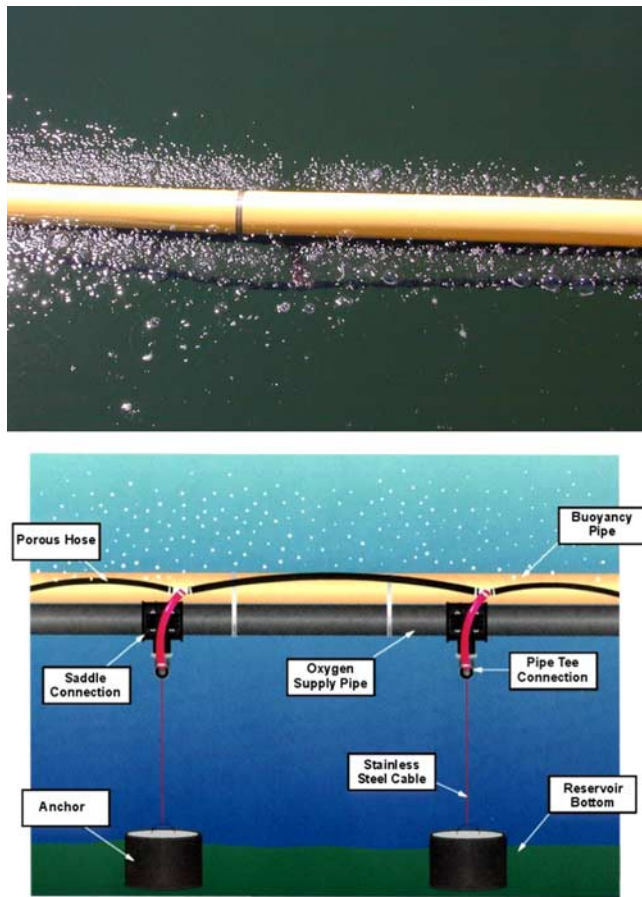


Figure 2. Photograph and schematic of linear bubble plume diffuser in Spring Hollow Reservoir, Virginia. Courtesy of Mark Mobley, Mobley Engineering, Inc.

operation, particularly for 2003 (Figures 4a–4d). Also, use of compressed air did not produce a strong DO plume signature for the July test compared with the August and October tests with pure oxygen. The actual plume rise height is easier to distinguish, especially in the DO contours for August and October. The predicted depths of maximum plume rise are 38.2, 46.1, and 48.8 m for July, August, and October, respectively. For July and October, the DMPR is simulated well by the model (Figures 4a, 4b, 4e, and 4f). However, the model appears to underestimate the plume rise height for August, when the gas flow rate was comparatively low (Table 3). The under-predicted DMPR may have been due to an overestimated value for α . In a detailed study of round plumes, *Milgram* [1983] found that α is directly proportional to the plume gas holdup or fraction. However, the linear bubble-plume model assumes that α is constant (Table 3).

[17] The structure of the plumes is similar to those observed by *Asaeda and Imberger* [1993] for round bubble plumes in weak stratification (Figure 4). Depending on the gas flow rate and stratification strength, three types of horizontal intrusions from the plume were reported. The pattern of the DO contours for October (Figure 4f) closely resembles type 1, which corresponds to a high gas flow rate or weak stratification [*Asaeda and Imberger*, 1993]. This is similar to a single plume impinging on a free surface, which

is analogous to the thermocline in SHR. Although the plume for July is not easily discerned from the temperature or DO contours, the structure is most likely similar to a type 1 plume. The July plume appears to detrain primarily at the lower thermocline with one strong intrusion, as evidenced by accumulation of higher oxygenated water near the top of the plume (Figure 4b). The plume for August is best classified as type 3, which is for low gas flow rates or strong density stratification [*Asaeda and Imberger*, 1993]. Type 3 plumes do not have steady intrusions, but instead are characterized by alternating, collapsing eddies that cause the plume to meander.

[18] Referring to the October test (Figure 4f), the higher DO concentrations at lower depths adjacent to both sides of the plume were likely the result of detrained water that sinks past the equilibrium depth due to momentum. The temperature isotherms were also depressed immediately beside the plume (Figure 4e). This phenomenon was also observed by *McGinnis et al.* [2004]. The DO concentration immediately above the predicted DMPR and near the vertical plume centerline for October 2004 is higher than the ambient concentration (Figure 4f). This could have been caused by the formation of a secondary plume above the DMPR resulting from incompletely dissolved bubbles. The model estimates that the bubble size at the top of the first plume was about 5×10^{-5} m for October 2004, which is relatively large compared with July 2003 and August 2003 (Figure 6f). These undissolved bubbles could have created a secondary plume, which entrained oxygenated water from the detrainment of the first plume and carried it higher into the water column.

[19] Vertical profiles of constituents and properties within the plumes were also predicted for the three diffuser tests (Figure 6). For the July test with compressed air, the higher gas flux creates a greater buoyancy flux and a higher initial plume water velocity (Figure 6d). Also, the concentration driving force for oxygen transfer is lower compared with pure oxygen, which decreases the rate of bubble dissolution with depth (Figure 6f). These effects result in a higher DMPR for July compared with August (Figures 4 and 6). The model predictions for 2004 differ from those of 2003 because of the differing boundary conditions (Figure 5). The plume rise height for the October test with pure oxygen is less than that for August, even though the gas flux was more than tripled (Table 3). The ambient temperature, and hence density, stratification was stronger in 2004, which provided greater negative buoyancy to decrease plume momentum. The stronger ambient density stratification also caused the plume velocity to decrease more rapidly with depth in 2004 despite a higher initial velocity from the diffuser (Figure 6d). The lower plume rise height in October resulted in lower plume water flow rates than July (Figure 6e), even with comparable gas fluxes applied (Table 3).

[20] Average temperature, DO, and density profiles within the plumes were also measured (Figures 6a–6c). The temperature, and consequently density, predictions for July and October deviate from the measured profiles where the plumes reach the top of the hypolimnion (Figures 6a and 6c), or where the rate of plume spreading is greatest (Figure 4). The model underpredicts the final plume temperature by approximately 0.3° and 0.2°C for July and October, respectively. One reason for the discrepancy

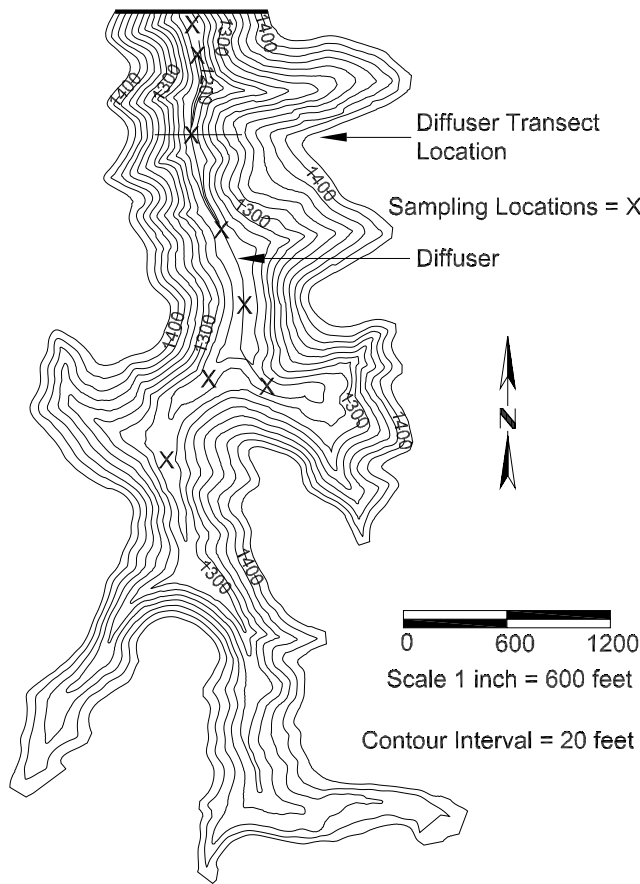


Figure 3. Bathymetric map of Spring Hollow Reservoir, Virginia, showing locations of linear bubble plume diffuser and conductivity-temperature-depth (CTD) lateral transects.

may be that the model assumes α is constant. In addition to the dependence on plume gas holdup, *Milgram* [1983] also found that α for a circular plume is directly proportional to the local plume radius. As the plume approaches its maximum rise height, the width increases rapidly

(Figure 4). If linear plume dynamics are similar to circular plumes, then α for linear plumes may also increase as the plume width increases. Additionally, the boundary profiles selected may not accurately reflect the actual ambient conditions immediately adjacent to the plume along its entire rise height. Measured vertical CTD profiles at the estimated plume width were averaged and used for boundary conditions. However, the plume width varies greatly with depth (Figure 4), so use of vertical profiles at a single lateral distance from the diffuser is not appropriate. Also, the average plume width was visually estimated from the temperature and DO contour plots (Figure 4), but the actual plume boundaries are not well defined.

[21] The model predicts the plume DO profiles well for all three diffuser tests (Figure 6b). For July and August the model characterizes the initial increase in DO immediately above the diffuser quite accurately. The initial rapid increase in DO for August and October is due to the higher oxygen saturation concentration at depth with the use of pure oxygen. The high hydrostatic pressure causes the oxygen transfer rate to be almost independent of the ambient DO concentration at the depth of diffuser. By contrast, the initial DO increase for the July test with compressed air is more modest (Figure 6b). The shape of the predicted DO profile for October differs somewhat from the experimental data, even though the final values at the DMPR differ by only 0.3 g/m³. The hypolimnion and hence the plume near-field were more heterogeneous in 2004 than 2003, which may have contributed to the overprediction at lower depths if the selected boundary profiles did not accurately represent water entrained into the plume.

[22] Another source of inaccuracy could be the correlation equation used to calculate initial bubble size (equation (3)). This relationship was developed using data collected over actual air flow rates per unit length of diffuser of 0.08–0.88 m²/h [*McGinnis and Little*, 2002]. The actual gas flow rates per unit length of diffuser for the July, August, and October tests were 0.019, 0.0063, and 0.024 m²/h, respectively. Therefore the initial bubble diameters used in the model were extrapolated beyond the valid correlation range.

Table 3. Conditions for Linear Bubble Plume Model Validation and Sensitivity Analysis for Spring Hollow Reservoir, Virginia^a

Parameter	2 July 2003	17 Aug 2003	23 Oct 2004	Sensitivity Range
Oxygen in gas supply, %	21	97	97	n/a
Entrainment coefficient []	0.11	0.11	0.11	0.05–0.2
Spreading coefficient []	0.93	0.93	0.93	0.5–1.0
Initial Froude number []	2.0	2.0	2.0	1.0–3.0
Operational diffuser length, m	300	300	250	60–625
Initial plume width, m	0.16	0.16	0.16	n/a
Initial plume area, m ²	50	50	42	10–100
Gas flow rate, ^b Nm ³ /h	38	13	40	1–400
Initial gas flux, ^c m/h	0.76	0.26	0.96	0.02–9.5
Initial bubble radius, m	5.7×10^{-4}	5.6×10^{-4}	5.7×10^{-4}	10^{-4} – 10^{-1}
Diffuser depth, ^d m	60	60	59	n/a
Reservoir maximum depth, m	64	65	64	n/a
Reservoir surface area, 10 ⁶ m ²	0.53	0.53	0.53	n/a
Reservoir total volume, 10 ⁶ m ³	12	12	12	n/a

^aReservoir conditions on testing days are also included.

^bNm³ denotes 1 m³ of gas at 1 bar and 0°C.

^cSensitivity analysis range refers to varying gas flow rate while maintaining constant initial plume area.

^dDepth at location of lateral CTD transect.

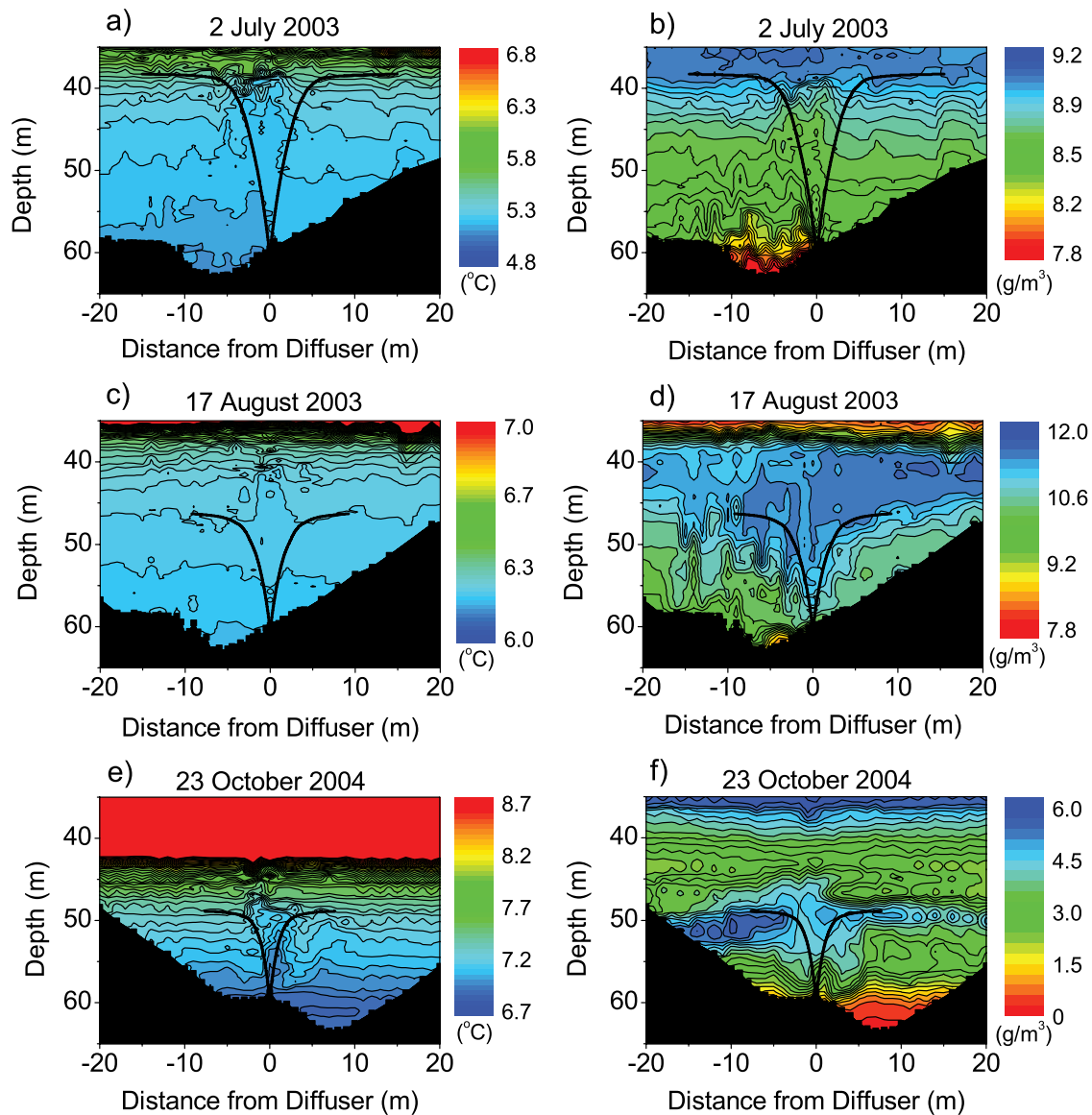


Figure 4. Measured (left) plume temperature ($^{\circ}\text{C}$) and (right) dissolved oxygen (DO) (g/m^3) contours with linear bubble plume model predictions for diffuser operation with air (2 July 2003) and pure oxygen (17 August 2003 and 23 October 2004) in Spring Hollow Reservoir, Virginia. Contours were interpolated from CTD profiles collected at locations indicated by small black squares along the bottom of each plot.

4.3. Sensitivity Analysis

[23] A sensitivity analysis was performed with the linear bubble plume model to determine the effects on plume rise height, oxygen transfer efficiency, and induced water flow rate, because these parameters are important for design and operation of hypolimnetic aeration/oxygenation systems. Parameter perturbation was used, in which input variables are independently adjusted to determine their individual effects on model predictions. The model variables investigated are either difficult to measure or can be controlled through system design or operation (Table 3). Currently, α and λ for the linear plume model are empirical constants [Fanneløp *et al.*, 1991], and the initial plume water velocity is calculated using a densimetric Fr [Wüest *et al.*, 1992]. The initial plume area, gas flow rate, and, to a lesser extent, initial bubble size can be controlled through diffuser design and operation. The sensitivity of model predictions to

ambient dissolved nitrogen was also examined for the standard case using compressed air (2 July 2003). The model assumes that the background dissolved nitrogen concentration is equivalent to the saturated value at atmospheric partial pressure and the average hypolimnetic water temperature.

[24] The DMPR is most influenced by gas flow rate and initial bubble radius (Figures 7e and 7f). The gas flow rate is directly related to the density of the plume bubble-water mixture and subsequently the positive buoyancy and upward momentum of the plume (Table 2). For gas flow rates greater than approximately $100 \text{ Nm}^3/\text{h}$, the plume rise height is controlled more by the depth of the thermocline in SHR (Figure 5a). As the plume rise height approaches the thermocline, further increases in the buoyancy and momentum fluxes can not overcome the strong ambient density stratification. The 2003 predictions are more a function of

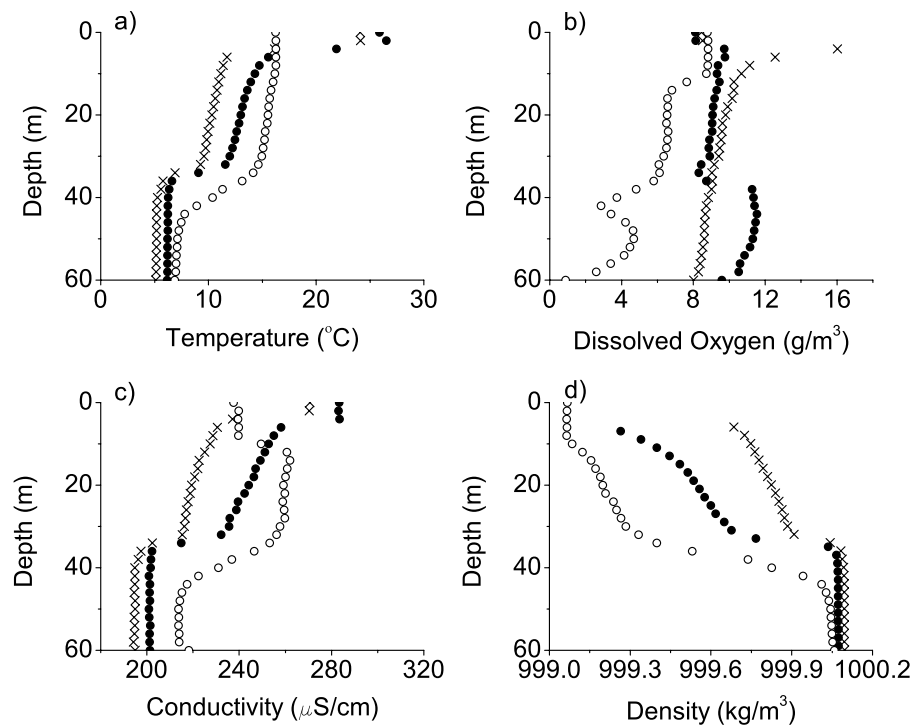


Figure 5. Input boundary conditions for linear bubble plume model validation and sensitivity analysis. Data collected from Spring Hollow Reservoir, Virginia, during diffuser operation with compressed air (2 July 2003, crosses) and pure oxygen (17 August 2003, solid circles, and 23 October 2004, open circles).

gas flow rate than those for October 2004 because of differing boundary conditions (Figure 5).

[25] The plume rise height is moderately sensitive to initial bubble radius (Figure 7f). For initial radii less than about 1 mm, the DMPRs are nearly independent of bubble size. As initial bubble sizes increase, the plumes ascend higher and reach a maximum height for a radius of about 6 mm for each diffuser test. The shapes of the curves can be attributed to the dependence of bubble rise velocity and gas transfer coefficients on bubble radius [Wüest *et al.*, 1992]. The DMPR predictions for August are more sensitive to initial bubble radius than those for July and October (Figure 7f). The plume on 17 August 2003 did not have sufficient buoyancy and momentum to rise to the thermocline because a relatively low gas flow rate was applied (Figure 4c and Table 3). By contrast, the plume dynamics for July and October were influenced to a greater degree by the thermocline, which in effect damped the sensitivity of plume rise height to the initial bubble radius.

[26] The predicted DMPR for the linear bubble plume is virtually independent of Fr_o and λ over the ranges analyzed (Figures 7b and 7c). The plume rise height is moderately sensitive to α . Entrainment into the plume is a function of plume size, water velocity, and α (Table 1), and entrainment of ambient water decelerates the plume. Plume rise is influenced to a somewhat greater degree by the initial plume area. As the plume area is increased, the buoyancy flux decreases because the gas flow rate is constant. Plume rise height was unaffected by variations in the ambient dissolved nitrogen concentration from 50 to 200% saturation in the hypolimnion for the 2 July 2003 diffuser test. Similar to the analysis for DMPR, induced water flow rate at the top of the plume was found to be insensitive to Fr_o and λ and

somewhat more sensitive to α and the initial plume area (results not shown). Plume water flow rate is most influenced by gas flow rate and initial bubble radius. Higher gas flow rates produce greater buoyancy and momentum fluxes, which results in greater plume rise heights and increased entrainment.

[27] Oxygen transfer efficiency (total mass of oxygen transferred relative to initial mass of oxygen in bubbles) within the hypolimnion was independent of all parameters except initial bubble radius, decreasing from nearly 100% to around 20% for air and pure oxygen as the initial radii increased from approximately 1 mm to 1 cm (Figure 8). Oxygen transfer can continue above the DMPR if the bubbles are not dissolved. However, secondary plumes are not accounted for in the model. Even though undissolved bubbles at the top of the plume may continue to transfer oxygen during ascent, the oxygen may not be added at the desired depth (i.e., below the thermocline). A local minimum with respect to induced water flow rate at the top of the plume is predicted for a bubble radius of about 1 mm for the conditions in SHR (not shown). This suggests that while maximum oxygen transfer efficiency can be achieved with a 1-mm initial bubble radius, vertical water circulation, and hence oxygen distribution in the hypolimnion, will not be optimized. As the bubble radius is increased from 1 mm, induced water flow rate increases as well for radii up to 5 mm, but oxygen transfer efficiency decreases rapidly within this range (Figure 8).

5. Comparison of Linear and Circular Bubble Plume Models

[28] The primary difference between the linear and circular bubble plume models is the plume geometry. For a

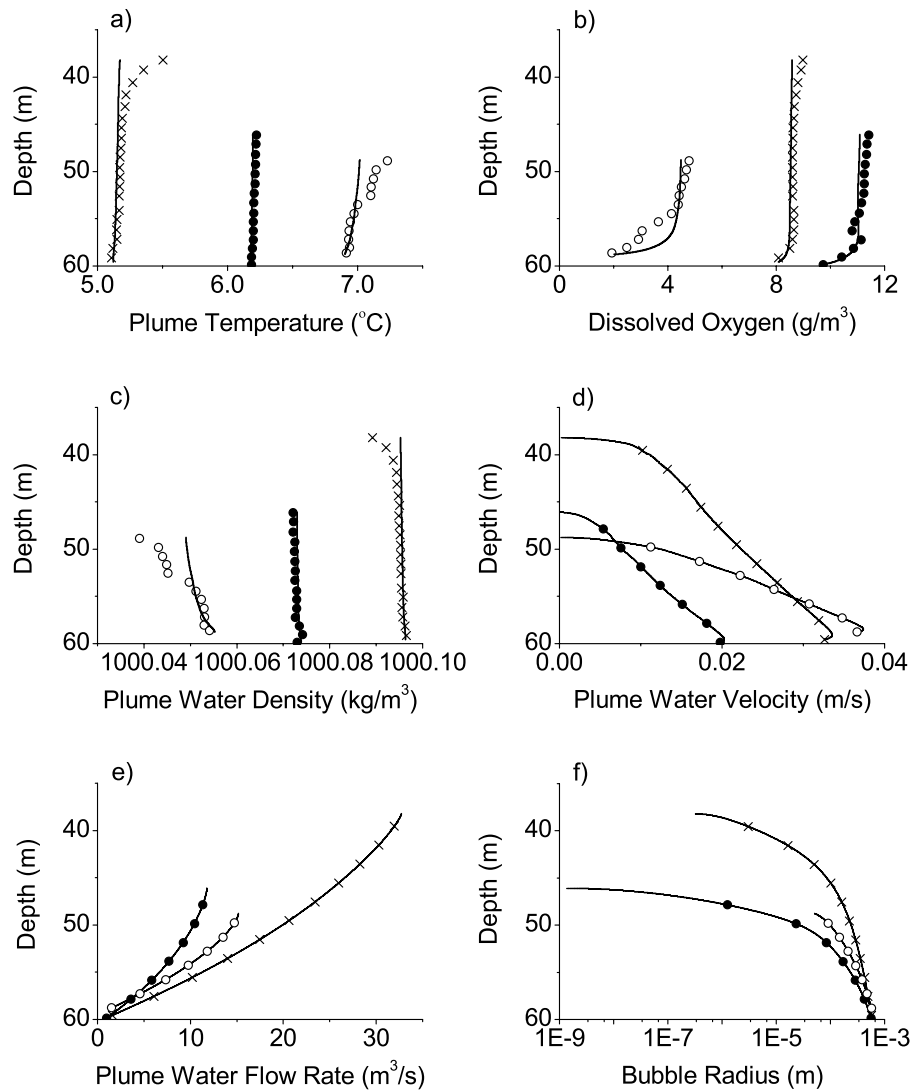


Figure 6. In-plume profiles predicted by linear bubble plume model, represented as (a–c) solid lines and (d–f) solid lines and symbols. Input data collected from Spring Hollow Reservoir, Virginia, during diffuser operation with compressed air (2 July 2003, crosses) and pure oxygen (17 August 2003, solid circles, and 23 October 2004, open circles). Measured average in-plume temperature, DO, and plume water density represented as symbols (Figures 6a–6c).

given plume cross-sectional area, the perimeter of a linear plume is much greater than for a circular or round plume. The initial estimated plume area and perimeter of the linear diffuser in SHR for July 2003 was about 50 m² and 600 m, respectively (Table 3). This corresponds to an equivalent radius and perimeter of approximately 4 m and 25 m, respectively, for a circular plume. In this case, the perimeter of the linear plume is 24 times greater than for the circular plume. For both the linear and circular plume models, entrainment of ambient water is directly proportional to local plume perimeter, local plume water velocity, and α (Table 1). The larger perimeter of the linear plume greatly increases ambient entrainment, which contributes to the negative buoyancy of the plume and causes the plume to decelerate more rapidly. This results in a lower plume rise height compared with the circular plume.

[29] The sensitivity of the linear bubble plume model to various parameters was comparable to results for the circu-

lar bubble plume model analysis. Gas flow rate had the greatest effect on DMPP predictions by both models, and the linear plume model was less sensitive to initial bubble radius than the circular plume model was [Wüest *et al.*, 1992]. The latter may be due to differences between temperature boundary conditions, which caused plume dynamics in SHR to be more influenced by thermocline depth. Initial bubble size greatly affected oxygen transfer efficiency for both diffuser geometries, decreasing rapidly as the radii increased beyond 1 mm and 3 mm for the linear and circular plumes, respectively. Both the linear and circular plume model predictions for DMPP were relatively insensitive to Fr_o and initial plume area (Figure 7 and Wüest *et al.* [1992]). However, circular model predictions for DMPP were more sensitive to α . Overall, the linear model was less sensitive to input parameters than the circular model, but this insensitivity is probably due to the relatively

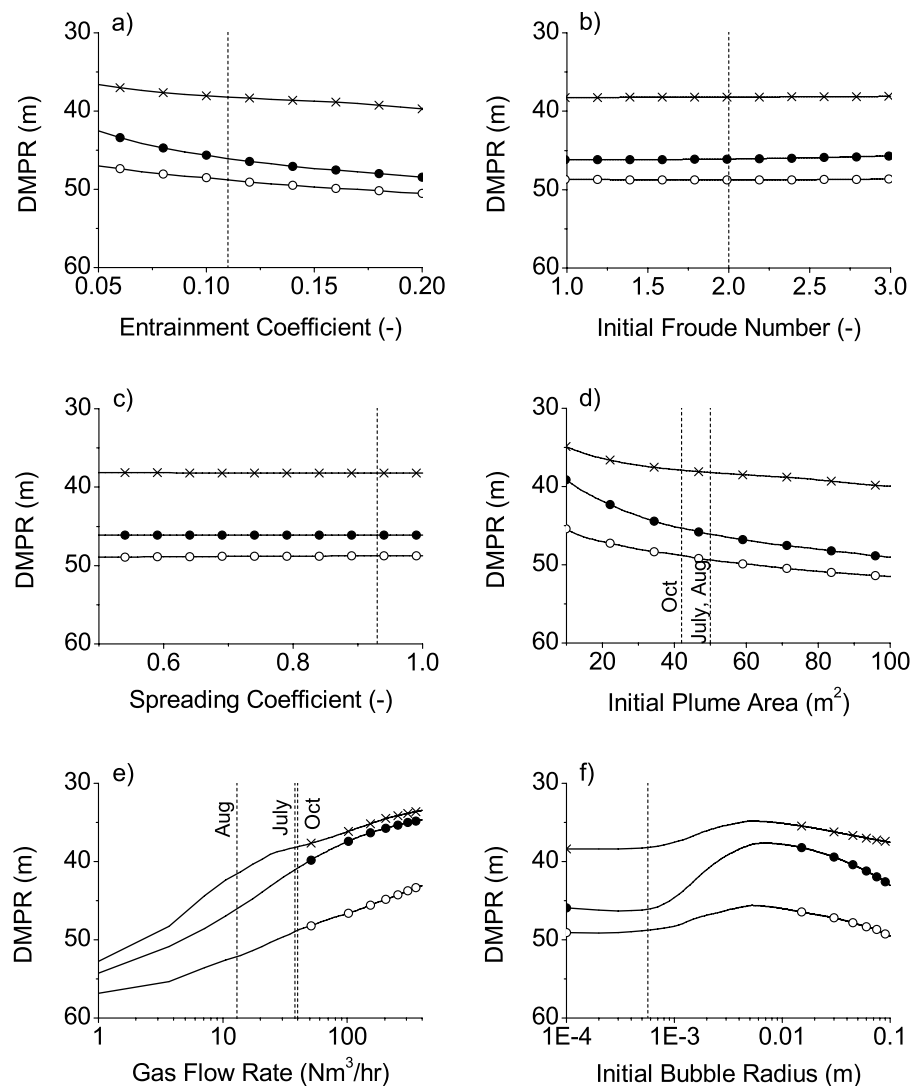


Figure 7. Effect of linear bubble plume model parameters on depth of maximum plume rise (DMPR). Standard values for each parameter are indicated by the vertical dashed lines. Input data collected from Spring Hollow Reservoir, Virginia, during diffuser operation with compressed air (2 July 2003, crosses) and pure oxygen (17 August 2003, solid circles, and 23 October 2004, open circles).

homogeneous boundary conditions in the hypolimnion of SHR caused by diffuser mixing.

6. Summary and Conclusions

[30] In the current work, the linear bubble plume model of McGinnis *et al.* [2001] was improved, and the updated model was evaluated using data collected from a full-scale hypolimnetic oxygenation system installed in Spring Hollow Reservoir, Virginia. Three diffuser experiments were conducted using compressed air and pure oxygen over a range of flow rates. Predicted plume rise height, spreading, and constituent profiles were compared with experimental observations. For July 2003 and October 2004, the DMPR was simulated well by the model. However, the model underestimated the plume rise height for August 2003, when the gas flow rate was comparatively low. The model underpredicted the final plume temperature by approximately 0.3° and 0.2°C for July 2003 and October 2004, respectively. The model predicted the plume DO

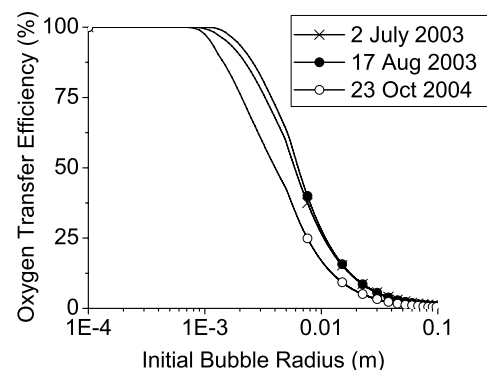


Figure 8. Effect of initial bubble radius on oxygen transfer efficiency predicted by linear bubble plume model. Input data collected from Spring Hollow Reservoir, Virginia, during diffuser operation with compressed air (2 July 2003) and pure oxygen (17 August 2003 and 23 October 2004).

profiles very well for all three diffuser tests, including simulating the initial rapid transfer of oxygen immediately above the diffuser.

[31] A sensitivity analysis was performed to determine the effect of various model input parameters. The DMPR and induced water flow rate were most influenced by gas flow rate and initial bubble radius, moderately sensitive to α and the initial plume area, and insensitive to Fr_o , λ , and ambient dissolved nitrogen. Oxygen transfer within the hypolimnion was independent of all parameters except initial bubble radius, decreasing from nearly 100% to around 20% for air and pure oxygen as the initial radii increased from approximately 1 mm to 1 cm.

[32] The linear bubble plume model for hypolimnetic oxygenation has been successfully validated. The model can be used to design lake and reservoir oxygenation systems and to optimize existing systems to maximize oxygen addition. Application of the linear plume model requires knowledge of the gas flow rate, initial bubble radius, initial plume area, and near-field constituent profiles (boundary conditions). Additionally, several empirical parameters must be estimated including α and λ . Even though all of the model inputs may not be known with certainty for a given aeration or oxygenation system, the model can be used for preliminary design and coarse optimization. Also, plume rise height, water flow rate, and oxygen transfer efficiency were found to be primarily dependent on gas flow rate and initial bubble radius.

[33] Operation of bubble plumes for hypolimnetic oxygenation usually alters the ambient temperature and DO conditions of a water body. Plume dynamics and oxygen transfer are strongly related to the near-field water column properties, establishing a feedback loop that continually changes plume dynamics. This complex plume-lake interaction should be accounted for in the design and operation of bubble plume diffusers. McGinnis *et al.* [2001] performed a preliminary coupling of the linear bubble plume model with an existing reservoir model, CE-QUAL-W2 [Cole and Wells, 2003], and obtained encouraging results. Efforts are currently under way to further develop the coupled model to predict plume performance and the near- and far-field reservoir responses. In the absence of near-field boundary profiles, far-field or simulated constituent profiles can be used to provide a reasonable estimate of plume performance.

Notation

b	plume radius (circular bubble plume), m.
d	bubble diameter, mm.
C	dissolved concentration, mol/m ³ .
E	entrainment factor, m ² /s.
F_D	dissolved species flux, mol/s.
F_G	gaseous species flux, mol/s.
F_S	salinity flux, kg/s.
F_T	temperature flux, °C m ³ /s.
Fr	Froude number [].
g	gravitational acceleration, m/s ² .
H	Henry's constant, mol/m ³ /bar.
K_L	mass transfer coefficient, m/s.
L	plume length, m.
M	water momentum, m ⁴ /s ² .

N	number flux of bubbles, 1/s.
P	pressure, bar.
Q	plume flow rate, m ³ /s.
q	actual gas flow rate per unit diffuser length, m ² /h.
Ri	Richardson number [].
R	bubble radius, m.
S	salinity, g/kg.
T	temperature, °C.
v	velocity, m/s.
W	plume width, m.
y	gaseous concentration, mol/m ³ .
Z	depth, m.

Greek letters

α	entrainment coefficient [].
λ	spreading coefficient [].
ρ	density, kg/m ³ .

Subscripts

3,2	Sauter-mean.
G	Gaussian profile.
O	oxygen.
N	nitrogen.
T	top-hat profile.
a	ambient water.
b	bubble.
i	gas species, oxygen or nitrogen.
o	initial.
p	plume water and gas mixture.
w	plume water.

[34] **Acknowledgments.** The authors thank Daniel McGinnis for his valuable technical assistance. Financial support was generously provided by the U.S. National Science Foundation (grant BES 0202034) and the Western Virginia Water Authority.

References

- Asaeda, T., and J. Imberger (1993), Structure of bubble plumes in linearly stratified environments, *J. Fluid Mech.*, 249, 35–57.
- Brevik, I. (1977), Two-dimensional air-bubble plume, *J. Waterw. Port Coastal Ocean Div. Am. Soc. Civ. Eng.*, 103, 101–115.
- Brevik, I., and R. Kluge (1999), On the role of turbulence in the phenomenological theory of plane and axisymmetric air-bubble plumes, *Int. J. Multiphase Flow*, 25, 87–108.
- Burris, V. L., D. F. McGinnis, and J. C. Little (2002), Predicting oxygen transfer and water flow rate in airlift aerators, *Water Res.*, 36, 4605–4615.
- Cole, T. M., and S. A. Wells (2003), CE-QUAL-W2: A two-dimensional, laterally averaged, hydrodynamic and water quality model, Version 3.2, U.S. Army Eng. and Res. Dev. Cent., Vicksburg, Miss.
- Cooke, G. D., and R. E. Carlson (1989), *Reservoir Management for Water Quality and THM Precursor Control*, 387 pp., Am. Water Works Assoc. Res. Found., Denver, Colo.
- Ditmars, J. D., and K. Cederwall (1974), Analysis of air-bubble plumes, paper presented at the 14th Coastal Engineering Conference, Am. Soc. of Civ. Eng., Copenhagen, 24–28 June.
- Fanneløp, T. K., and K. Sjøen (1980), Hydrodynamics of underwater blowouts, *Norw. Marit. Res.*, 4, 17–33.
- Fanneløp, T. K., and D. M. Webber (2003), On buoyant plumes rising from area sources in a calm environment, *J. Fluid Mech.*, 497, 319–344.
- Fanneløp, T. K., S. Hirschberg, and J. Kueffer (1991), Surface current and recirculating cells generated by bubble curtains and jets, *J. Fluid Mech.*, 229, 629–657.
- Fischer, H. B., E. J. List, R. C. Y. Koh, J. Imberger, and N. H. Brooks (1979), *Mixing in Inland and Coastal Waters*, 483 pp., Elsevier, New York.
- Johansen, O. (2000), DeepBlow—A Lagrangian plume model for deep water blowouts, *Spill Sci. Technol. Bull.*, 6, 103–111.
- Kobus, H. E. (1968), Analysis of the flow induced by air-bubble systems, paper presented at the 11th Coastal Engineering Conference, Am. Soc. of Civ. Eng., London.

- Laureshen, C. J., and R. D. Rowe (1987), Modeling of plane bubble plumes, paper presented at the 24th National Heat Transfer Conference and Exhibition, Am. Soc. of Mech. Eng., Pittsburgh, Pa., 9–12 Aug.
- Lemckert, C. J., and J. Imberger (1993), Energetic bubble plumes in arbitrary stratification, *J. Hydraul. Eng.*, 119, 680–703.
- McDougall, T. J. (1978), Bubble plumes in stratified environments, *J. Fluid Mech.*, 85, 655–672.
- McGinnis, D. F., and J. C. Little (1998), Bubble dynamics and oxygen transfer in a Speece cone, *Water Sci. Technol.*, 37, 285–292.
- McGinnis, D. F., and J. C. Little (2002), Predicting diffused-bubble oxygen transfer rate using the discrete-bubble model, *Water Res.*, 36, 4627–4635.
- McGinnis, D. F., J. C. Little, and A. Wuest (2001), Hypolimnetic oxygenation: Coupling bubble-plume and reservoir models, paper presented at Asian Waterqual 2001: First IWA Asia-Pacific Regional Conference, Int. Water Assoc., Fukuoka, Japan, 12–15 Sept.
- McGinnis, D. F., A. Lorke, A. Wüest, A. Stöckli, and J. C. Little (2004), Interaction between a bubble plume and the near field in a stratified lake, *Water Resour. Res.*, 40, W10206, doi:10.1029/2004WR003038.
- Milgram, J. H. (1983), Mean flow in round bubble plumes, *J. Fluid Mech.*, 133, 345–376.
- Peterson, M. J., G. F. Cada, M. J. Sale, and G. K. Eddlemon (2003), Regulatory approaches for addressing dissolved oxygen concerns at hydropower facilities, 38 pp., U.S. Dep. of Energy Off. of Energy Efficiency and Renewable Energy, Idaho Falls, Idaho.
- Rayyan, F., and R. E. Speece (1977), Hydrodynamics of bubble plumes and oxygen absorption in stratified impoundments, *Prog. Water Technol.*, 9, 129–142.
- Sahoo, G. B., and D. Luketina (2003), Modeling of bubble plume design and oxygen transfer for reservoir restoration, *Water Res.*, 37, 393–401.
- Schladow, S. G. (1992), Bubble plume dynamics in a stratified medium and the implications for water quality amelioration in lakes, *Water Resour. Res.*, 28, 313–321.
- Shang, E. H. H., R. M. K. Yu, and R. S. S. Wu (2006), Hypoxia affects sex differentiation and development, leading to a male-dominated population in zebrafish (*Danio rerio*), *Environ. Sci. Technol.*, 40, 3118–3122.
- Singleton, V. L., and J. C. Little (2005), Linear bubble plume model for hypolimnetic oxygenation: Full-scale evaluation and sensitivity analysis, paper presented at 9th Workshop on Physical Processes in Natural Waters, Lancaster Univ., Lancaster, UK, 4–6 Sept.
- Singleton, V. L., and J. C. Little (2006), Designing hypolimnetic aeration and oxygenation systems: A review, *Environ. Sci. Technol.*, 40(24), 7512–7520, doi:10.1021/es060069s.
- Wilkinson, D. L. (1979), Two-dimensional bubble plumes, *J. Hydraul. Div. Am. Soc. Civ. Eng.*, 105, 139–154.
- Wüest, A., N. H. Brooks, and D. M. Imboden (1992), Bubble plume modeling for lake restoration, *Water Resour. Res.*, 28, 3235–3250.
- Zheng, L., P. D. Yapa, and F. Chen (2002), A model for simulating deep-water oil and gas blowouts: I. Theory and model formulation, *J. Hydraul. Res.*, 41, 339–351.

P. Gantzer, J. C. Little, and V. L. Singleton, Department of Civil and Environmental Engineering, Virginia Polytechnic Institute and State University, 418 Durham Hall, Blacksburg, VA 24061-0246, USA. (jcl@vt.edu)